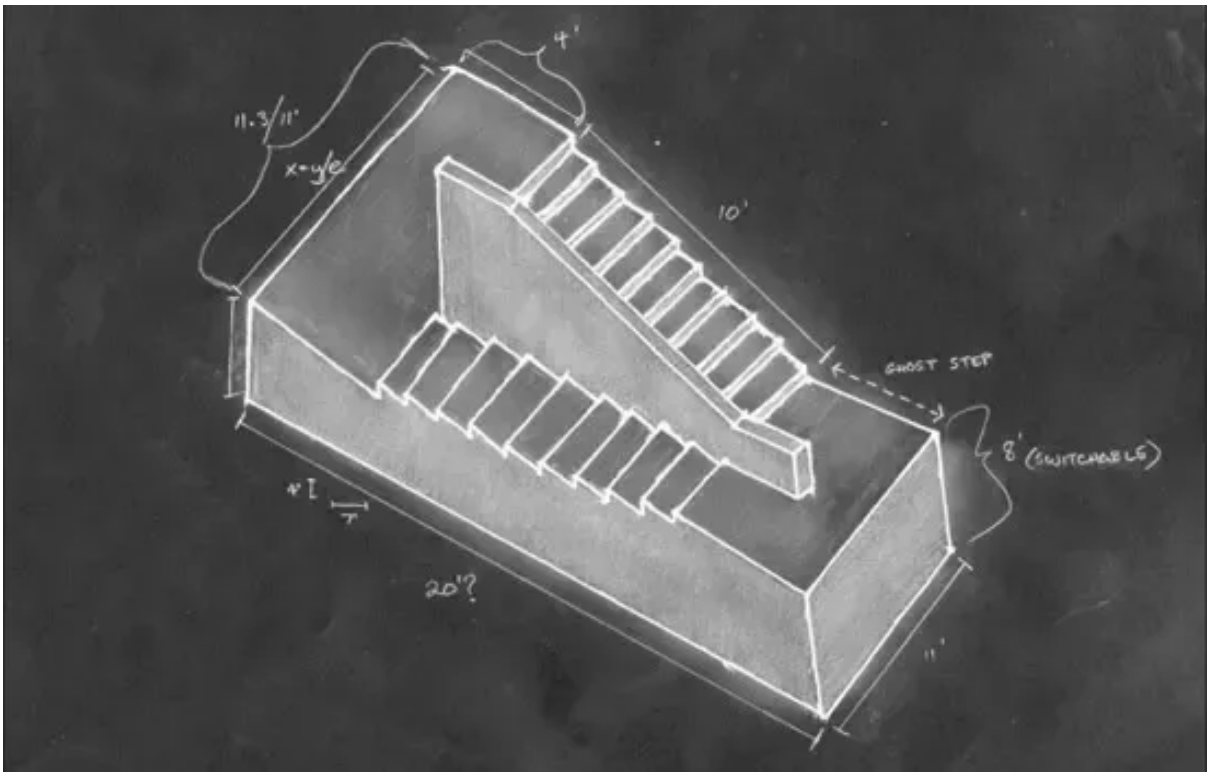
Cthulhu's cat sitter tech notes

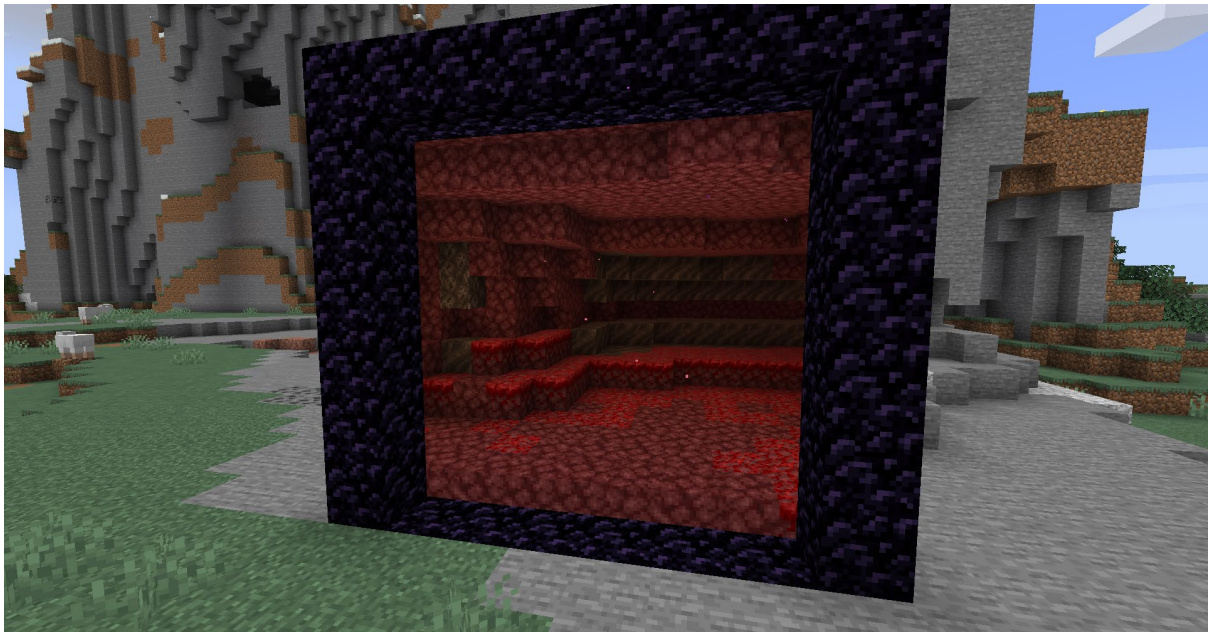
- Techniques foe non-Euclidean mechanics
- Reference material for style
- Blender basics
- Working between programs



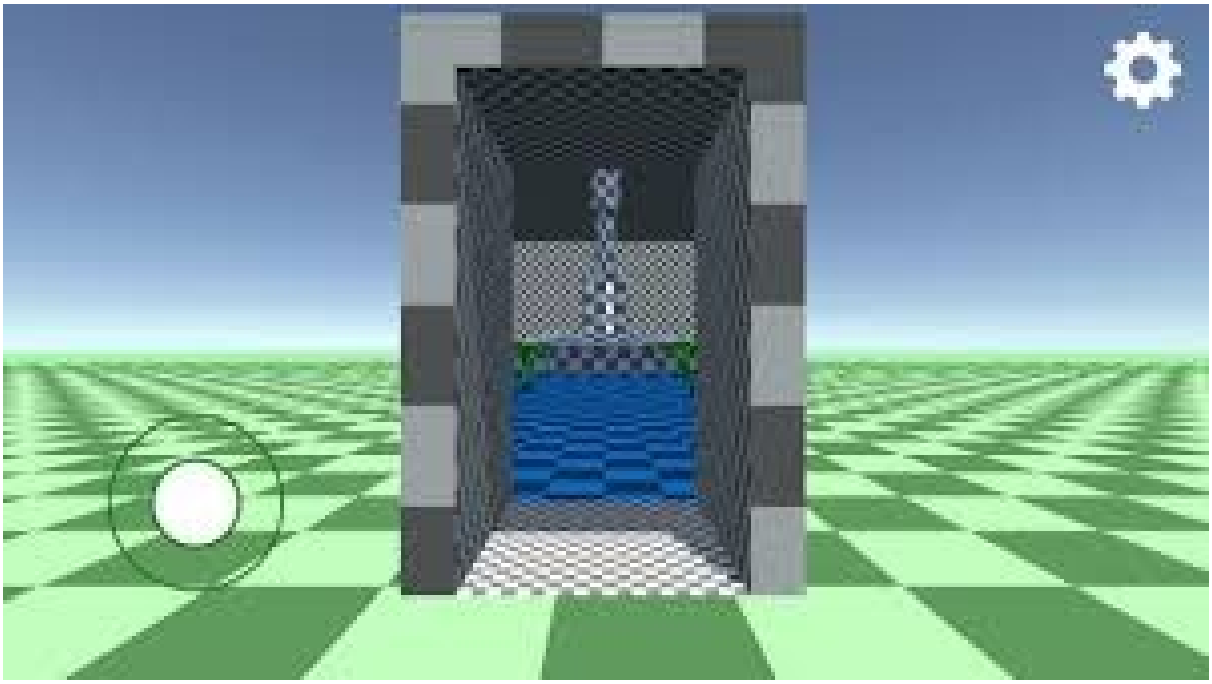
Utilising a game engine, we can create a perfect infinite staircase by creating a set of stairs that appear the same on the top landing as on the bottom landing, when the player reaches the top of the stairs they are instantly teleported to the bottom landing in the corresponding position. By playing with this further you could create a staircase that requires you to go up twice then down again in order to go up a floor. This doesn't make sense in the real world but in the game, it would behave seamlessly creating a working non-Euclidean environment that the player would have to work out how to navigate effectively. A good simple trick would be to construct a spiral staircase that takes more rotation to go up for example than it would take to go down. Whilst it isn't hard to navigate it would hopefully act to disorient the player a bit.

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Portals are also non-Euclidean.



I am a particular fan of this mod developed for Minecraft ^^^



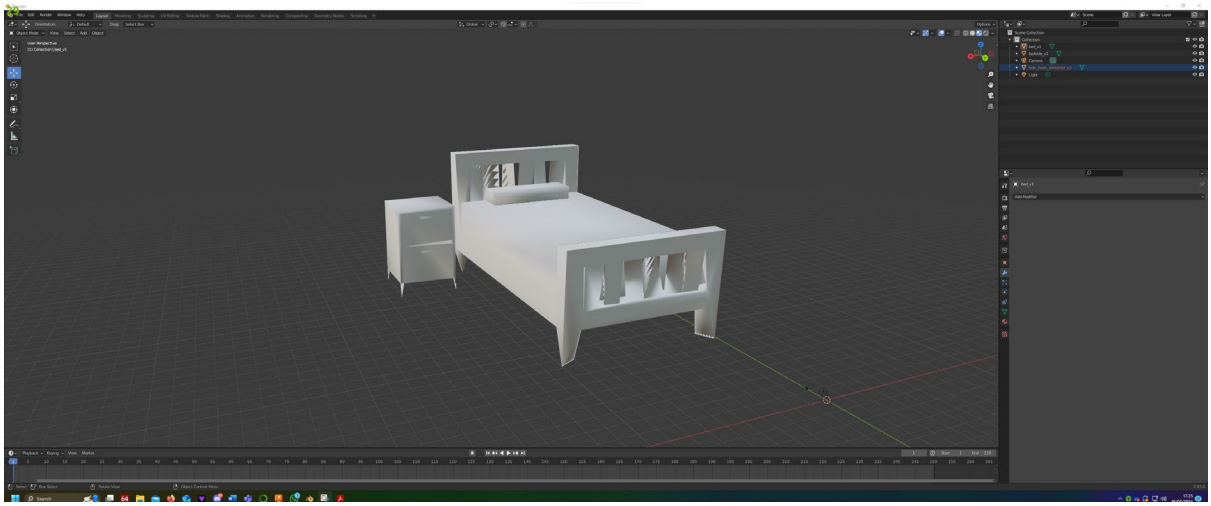
Utilising a portal it is very easy to create the Tardis effect, simply make the interior of the Tardis exist somewhere outside and then when you walk through the doors, teleport to the interior. The same is done in reverse to exit of course.

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I had a play around in blender and experimented with using modifiers on solid body models that I made in a different program. What I discovered is that if the level of detail is kept relatively uniform throughout a model, a simple decimate modifier can be used to create a fairly presentable low Poly version of the model. To me, low poly modelling is a very foreign concept. My 3D modelling experience is derived from engineering-based CAD programmes Which deal with solid bodies instead of polygons and where the aim is to model the components as accurately as possible. In this system a curved surface is a curved surface rather than an approximation made-up of triangles. This distinction is what makes it so unnatural for me. For example, to model a glass I would probably start by drawing a cross section and revolving it around the central axis. If I wish to add decoration to it, I would simply modify the surface of the glass and duplicate the pattern, rotating it around the same centre axis. To create a low Poly model of a glass, you wouldn't even consider a circle.

Whilst it's true that you can utilise a process using various 2D reference images to create a boxy approximation of a 3D character/model, my lack of familiarity with blender makes this difficult. As a result, I have decided to work on my blender skills in parallel with generating assets within fusion 360. I believe this hybrid approach will allow me to create models efficiently whilst still developing my skill set.

The assets produced from decimating solid body models are distinctive as they never simplify fully uniformly. Personally, I don't see this as a bad thing, if the vast majority of the assets conform to this pattern, then the styling is in keeping and matches the desired theme very well. In fact, I believe that the slightly "Janky" low Poly models will help bridge the gap between the modern game engine use the developer project and the original PlayStation style graphics which we're trying to emulate.



When it comes to deciding upon a file type to transfer models between different programmes, I have several options. The main decision I have to make is whether I export a mesh from fusion 360 and import them mesh straight into blender or to export a solid model and perform the conversion externally before importing into blender. Fusion has gotten pretty good at exporting meshes so I'm confident that exporting a mesh is the best choice. This then begs the question of which file format to choose. I chose .obj as a format as it offers some benefits over. stl's in terms of quality but also doesn't have the extra unused data that would come with an. fbx. These concerns aren't really that necessary but I feel it's best to maintain good practice where not inconvenient to do so.

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